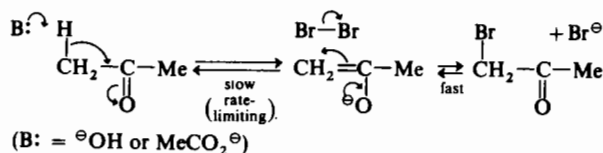


By analogy with acids above, specific basic catalysis is found to be characteristic of reactions in which there is rapid, reversible proton-removal from the substrate *before* the slow, rate-limiting step.

In *general base catalysis*, bases other than $^{\ominus}\text{OH}$ are involved. Thus in the base catalysed bromination of acetone (*cf.* p. 295) in an acetate buffer it is found that,

$$\text{Rate} = k_{^{\ominus}\text{OH}}[^{\ominus}\text{OH}][\text{MeCOMe}] + k_{\text{MeCO}_2^{\ominus}}[\text{MeCO}_2^{\ominus}][\text{MeCOMe}]$$

and the reaction is believed to proceed:



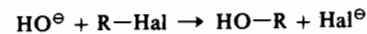
Again by analogy with acids above, general base catalysis is found to be characteristic of reactions in which removal of proton from the substrate is slow, i.e. rate-limiting, and is followed by rapid conversion of the intermediate into products.

4

Nucleophilic substitution at a saturated carbon atom

- 4.1 RELATION OF KINETICS TO MECHANISM, p. 77.
- 4.2 EFFECT OF SOLVENT, p. 80.
- 4.3 EFFECT OF STRUCTURE, p. 82.
- 4.4 STEREOCHEMICAL IMPLICATIONS OF MECHANISM, p. 87:
 - 4.4.1 $\text{S}_{\text{N}}2$ mechanism: inversion of configuration, p. 87;
 - 4.4.2 Determination of relative configuration, p. 88; 4.4.3 $\text{S}_{\text{N}}1$ mechanism: racemisation?, p. 90; 4.4.4 The mechanistic borderline, p. 91; 4.4.5 $\text{S}_{\text{N}}i$ mechanism: retention of configuration, p. 92; 4.4.6 Neighbouring group participation: 'retention', p. 93.
- 4.5 EFFECT OF ENTERING AND LEAVING GROUPS, p. 96:
 - 4.5.1 The entering group, p. 96; 4.5.2 The leaving group, p. 98.
- 4.6 OTHER NUCLEOPHILIC DISPLACEMENTS, p. 99.

A type of reaction that has probably received more detailed study than any other—largely due to the monumental work of Ingold and his school—is nucleophilic substitution at a saturated carbon atom: the classical displacement reaction exemplified by the conversion of an alkyl halide into an alcohol by the action of aqueous base:



Kinetic measurements on reactions in which alkyl halides are attacked by a wide variety of different nucleophiles, $\text{Nu}^{\ominus}$, have revealed two, essentially extreme, types: one in which,

$$\text{Rate} = k_2[\text{RHal}][\text{Nu}^{\ominus}] \quad [1]$$

and another in which,

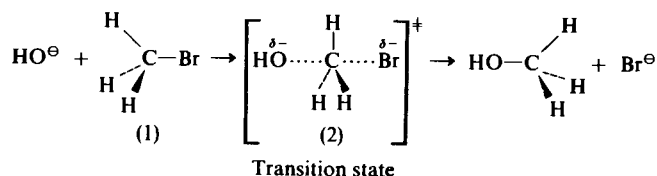
$$\text{Rate} = k_1[\text{RHal}] \quad [2]$$

i.e. the rate is independent of $[\text{Nu}^{\ominus}]$. In some cases the rate equations are found to be 'mixed' or are otherwise complicated, but examples are known which exactly follow the simple relations above.

4.1 RELATION OF KINETICS TO MECHANISM

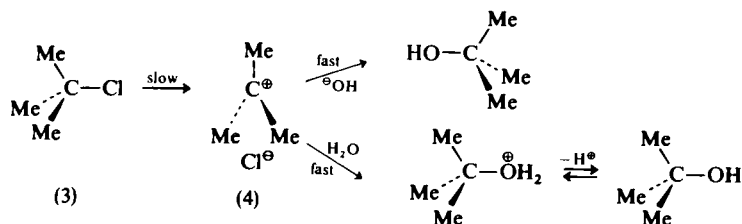
Hydrolysis of the primary halide bromomethane (methyl bromide) in aqueous base has been shown to proceed according to equation [1]

above, and this has been interpreted as involving the participation of both alkyl halide and hydroxyl ion in the rate-limiting (i.e. slowest) step of the reaction. Ingold has suggested a transition state in which the attacking hydroxyl ion becomes partially bonded to the reacting carbon atom before the incipient bromide ion has become wholly detached from it; thus part of the energy necessary to effect the breaking of the C—Br bond is then supplied by that produced in forming the HO—C bond. Quantum mechanical calculation shows that an approach by the hydroxyl ion along the line of centres of the carbon and bromine atoms is that of lowest energy requirement. This can be represented:



The negative charge is spread in the transition state in the course of being transferred from hydroxyl to bromine, and the hydrogen atoms attached to the carbon atom attacked pass through a position in which they all lie in one plane (at right angles to the plane of the paper as drawn above). The initially sp^3 hybridised carbon atom becomes sp^2 hybridised in the transition state, the HO and Br being associated with the two lobes of the unhybridised p orbital that is thereby made available. This type of mechanism has been designated by Ingold as S_N2 : Substitution Nucleophilic bimolecular.

By contrast, hydrolysis of the tertiary halide 2-chloro-2-methylpropane (3,*t*-butyl chloride) in base is found kinetically to follow equation [2], i.e. as the rate is independent of $[\text{OH}^\ominus]$, this can play no part in the rate-limiting step. This has been interpreted as indicating that the halide undergoes slow ionisation (in fact, completion of the $\text{R} \rightarrow \text{Cl}$ polarisation that has already been shown to be present in such a molecule) as the rate-limiting step to yield the ion pair $\text{R}^\oplus\text{Cl}^\ominus$ (4); followed by rapid, non rate-limiting attack by $\text{OH}^\ominus$ or, if that is suitable, by solvent, the latter often predominating because of its very high concentration:



This type of mechanism has been designated S_N1 : Substitution Nucleophilic unimolecular. The energy necessary to effect the initial ionisation is largely recovered from the energy evolved through solvation of the resultant ion pair. The entropy of activation, $\Delta S^\ddagger$, for such a dissociative process (cf. p. 39) is also advantageous; thus $\Delta S^\ddagger$ for the hydrolysis of Me_3CCl is found to be $+51 \text{ J K}^{-1} \text{ mol}^{-1}$, compared with $-17 \text{ J K}^{-1} \text{ mol}^{-1}$ for hydrolysis of CH_3Cl . The cation in the ion pair (4), in which the central carbon atom carries the +ve charge, is of course a carbocation intermediate, and during its formation the initially sp^3 hybridised carbon atom collapses to a more stable planar (sp^2) state, in which the three methyl groups are as far apart from each other as they can get. Attack by $\text{OH}^\ominus$ or solvent (e.g. H_2O) can then take place from either side of this planar intermediate. If attainment of this planar state is inhibited by steric or other factors (cf. p. 87), the carbocation intermediate will be formed only with difficulty, if at all; i.e. ionisation, and hence reaction by the S_N1 pathway, may then not take place.

Thus the salient difference between reaction by the S_N2 and S_N1 pathways is that S_N2 proceeds in one step only, *via* a transition state; while S_N1 proceeds in two steps, *via* an actual (carbocation) intermediate.

A certain element of confusion is to be met with both in textbooks, and in the literature, over the use and meaning of the terms *order* (cf. p. 39) and *molecularity* as applied to reactions. The order is an experimentally determined quantity, the overall order of a reaction being the sum of the powers of the concentration terms that appear in the rate equation:

$$\text{Rate} = k_3[\text{A}][\text{B}][\text{C}] \quad \text{Third order overall}$$

$$\text{Rate} = k_3[\text{A}]^2[\text{B}] \quad \text{Third order overall}$$

$$\text{Rate} = k_2[\text{A}]^2 \quad \text{Second order overall}$$

Generally, however, it is the order with respect to a particular reactant (or reactants) that is of more interest and significance than the overall order, i.e. that the above reactions are first order, second order, and second order, respectively, with respect to A. Examples of both zero order, and non-integral orders, with respect to a particular reactant are also known.

The molecularity refers to the number of species (molecules, ions, etc.) that are undergoing bond-breaking and/or bond-making in one step of the reaction, usually in the rate-limiting step. It is important to realise that the molecularity is not an experimentally determined quantity, and has significance only in the light of the particular mechanism chosen for the reaction: it is an integral part of the mechanistic interpretation of the reaction and is susceptible to re-

evaluation, in the light of additional experimental information about the reaction, in a way that the order cannot be. The molecularity of the reaction as a whole only has meaning if the reaction proceeds in a single step (an *elementary* reaction), as is believed to be the case with the hydrolysis of bromomethane above (p. 78); order and molecularity then coincide, the reaction being second order overall (first order in each reactant) and bimolecular. Order and molecularity do not always, or necessarily, have the same value, however.

Simple kinetic measurements can, however, be an inadequate guide to which of the above two mechanisms, S_N1 or S_N2 , is actually operating in, for example, the hydrolysis of a halide. Thus, as we have seen (p. 45), where the solvent can act as a nucleophile (*solvolysis*), e.g. H_2O , we would expect for an S_N2 type reaction,

$$\text{Rate} = k_2[R\text{Hal}][H_2O]$$

but as $[H_2O]$ remains effectively constant the rate equation actually observed will be,

$$\text{Rate} = k_{\text{obs}}[R\text{Hal}]$$

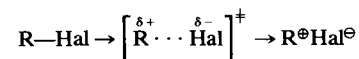
and simple kinetic measurements in aqueous solution will thus suggest, erroneously, that the reaction is of the S_N1 type.

A kinetic distinction between the operation of the S_N1 and S_N2 modes can often be made by observing the effect on the overall reaction rate of adding a competing nucleophile, e.g. azide anion, N_3^- . The total nucleophile concentration is thus increased, and for the S_N2 mode where $[Nu:]$ appears in the rate equation, this will result in an increased reaction rate due to the increased $[Nu:]$. By contrast, for the S_N1 mode $[Nu:]$ does not appear in the rate equation, i.e. is not involved in the rate-limiting step, and addition of N_3^- will thus be without significant effect on the observed reaction rate, though it will naturally influence the composition of the product.

4.2 EFFECT OF SOLVENT

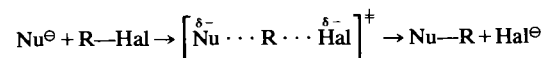
Changing the solvent in which a reaction is carried out often exerts a profound effect on its rate and may, indeed, even result in a change in its mechanistic pathway. Thus for a halide that undergoes hydrolysis by the S_N1 mode, increase in the polarity of the solvent (i.e. increase in ϵ , the dielectric constant) and/or its ion-solvating ability is found to result in a very marked increase in reaction rate. Thus the rate of solvolysis of the tertiary halide, Me_3CBr , is found to be 3×10^4 times faster in 50% aqueous ethanol than in ethanol alone. This occurs because, in the S_N1 mode, charge is developed and concentrated in

the T.S. compared with the starting material:



The energy required to effect such a process decreases as ϵ rises; the process is also facilitated by increasing solvation, and consequent stabilisation, of the developing ion pair compared with the starting material. That such effects, particularly solvation, are of prime importance is borne out by the fact that S_N1 type reactions are extremely uncommon in the gas phase.

For the S_N2 mode, however, increasing solvent polarity is found to have a much less marked effect, resulting in a slight decrease in reaction rate. This occurs because in this particular example new charge is not developed, and existing charge is dispersed, in the T.S. compared with the starting materials;



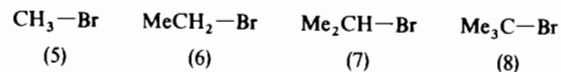
thus, solvation of the T.S. is likely to be somewhat less effective than that of the initial nucleophile—hence the slight decrease. This differing behaviour of S_N1 and S_N2 modes to changes of solvent can be used to some extent diagnostically.

A very marked effect on the rate of S_N2 reactions is, however, effected on transferring them from polar hydroxylic solvents to polar non-hydroxylic solvents. Thus the reaction rate of the primary halide, MeI , with N_3^- at 0° increased 4.5×10^4 -fold on transfer from methanol ($\epsilon = 33$) to N,N -dimethylmethanamide (dimethyl formamide, DMF), $HCONMe_2$, with very much the same polarity ($\epsilon = 37$). This very large rate difference stems from the fact that the attacking nucleophile, N_3^- , is highly solvated through hydrogen-bonding in $MeOH$ (cf. p. 57) whereas it is very much less strongly solvated—and not by hydrogen-bonding—in $HCONMe_2$. The largely unsolvated N_3^- anion (in $HCONMe_2$) is a very much more powerful nucleophile than when surrounded (as in $MeOH$) by a very much less nucleophilic solvation envelope, hence the rise in reaction rate. Rate increases of as much as 10^9 -fold have been observed on transferring S_N2 reactions from, e.g. $MeOH$, to another polar non-protic solvent, dimethyl sulphoxide (DMSO), Me_2SO ($\epsilon = 46$).

So far as actual changes of mechanistic pathway with change of solvent are concerned, increase in solvent polarity and ion-solvating ability may (but not necessarily will) change the reaction mode from $S_N2 \rightarrow S_N1$. Transfer from hydroxylic to polar, non-protic solvents (e.g. DMSO) can, and often do, change the reaction mode from $S_N1 \rightarrow S_N2$ by enormously increasing the effectiveness of the nucleophile in the system.

4.3 EFFECT OF STRUCTURE

An interesting sequence is provided by the reaction with base of the series of halides:



The first and last members are described in the literature as undergoing ready hydrolysis, the two intermediate members being more resistant. Measurement of their rates of hydrolysis with dilute, aqueous ethanolic sodium hydroxide solution gives the plot* (Fig. 4.1),

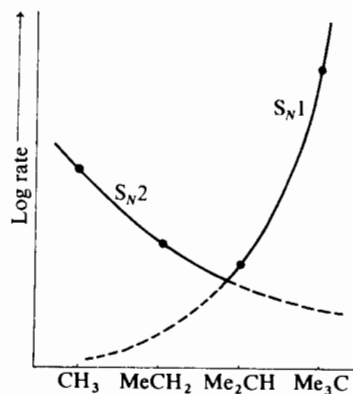


Fig. 4.1

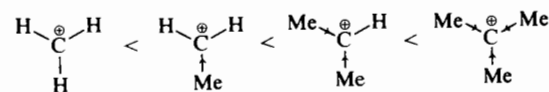
and further kinetic investigation reveals a change in order of reaction, and hence presumably, of mechanism, as the series is traversed. Thus bromomethane (5) and bromoethane (6) are found to follow a second order rate equation, 2-bromopropane (7) a mixed second and first order equation, the relative proportion of the two depending on the initial $[\text{OH}^-]$ (the higher the initial concentration the greater the second order proportion) and the total rate here being a minimum for the series, while 2-bromo-2-methylpropane (8) is found to follow a first order rate equation.

In seeking an explanation for the implied changeover in mechanistic pathway we need to consider, in each case, the effect on the transition state of both electronic and steric factors. For $\text{S}_{\text{N}}2$ attack, the enhanced inductive effect of an increasing number of methyl groups, as we go across the series, might be expected to make the carbon atom that

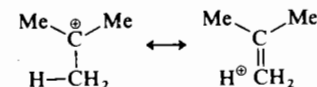
bears the bromine progressively less positively polarised, and hence less readily attacked by OH^- . This effect is probably small, and steric factors are of more importance; thus OH^- will find it progressively more difficult to attack the bromine-carrying carbon as the latter becomes more heavily substituted. More significantly, the resultant $\text{S}_{\text{N}}2$ transition state will have *five* groups around this carbon atom (compared with only *four* in the initial halide), there will thus be an increase in crowding on going from the initial halide to the transition state, and this relative crowding will increase as the size of the original substituents increases ($\text{H} \rightarrow \text{Me}$). The more crowded the T.S. relative to the starting materials, the higher its energy will be, and the slower therefore will it be formed. We would thus expect the purely $\text{S}_{\text{N}}2$ reaction rate to decrease as the above series is traversed. It is in fact possible to effect nucleophilic substitution ($\text{Br}^- + \text{R—Cl}$) on a series of halides analogous to those in Fig. 4.1 (p. 82), under conditions such that a strictly second order rate equation ($\text{S}_{\text{N}}2$ pathway) is followed throughout. We then observe:

Relative $\text{S}_{\text{N}}2$ rate:	CH_3Cl	MeCH_2Cl	Me_2CHCl	Me_3CCl
	1	2.7×10^{-2}	4.9×10^{-4}	2.2×10^{-5}

For $\text{S}_{\text{N}}1$ attack, considerable charge separation has taken place in the T.S. (cf. p. 81), and the ion pair intermediate to which it gives rise is therefore often taken as a model for it. As the above halide series is traversed, there is increasing stabilisation of the carbocation moiety of the ion pair, i.e. increasing rate of formation of the T.S. This increasing stabilisation arises from the operation of both an inductive effect,



and hyperconjugation (p. 25), e.g.



via the hydrogen atoms attached to the α -carbons, the above series of carbocations having 0, 3, 6 and 9 such hydrogen atoms, respectively.

Support for such an interaction of the H—C bonds with the carbon atom carrying the positive charge is provided by substituting H by D in the original halide, the rate of formation of the ion pair is then found to be slowed down by $\approx 10\%$ per deuterium atom incorporated: a result compatible only with the H—C bonds being involved in the ionisation. This is known as a *secondary kinetic isotope effect*, secondary

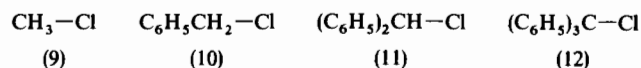
* Based on Ingold, *Structure and Mechanism in Organic Chemistry*, by permission of Cornell University Press.

because it is a bond other than that carrying the isotopic label that is being broken (*cf.* p. 46). The relative contributions of hyperconjugation and inductive effects to the stabilisation of carbocations is open to debate, but it is significant that a number of carbocations will only form at all if they can take up a planar arrangement, the state in which hyperconjugation will operate most effectively (*cf.* p. 104).

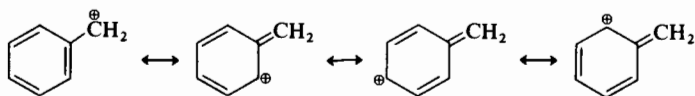
In steric terms there is a relief of crowding on going from the initial halide, with a tetrahedral disposition of four substituents about the sp^3 hybridised carbon atom, to the carbocation, with a planar disposition of only *three* substituents (*cf.* *five* for the S_N2 T.S.) about the now sp^2 hybridised carbon atom. The three substituents are as far apart from each other as they can get in the planar carbocation, and the relative relief of crowding (halide $\rightarrow$ carbocation) will increase as the substituents increase in size ($H \rightarrow Me \rightarrow Me_3C$). The S_N1 reaction rate would thus be expected to increase markedly (on both electronic and steric grounds) as the series of halides is traversed. It has not, however, proved possible to confirm this experimentally by setting up conditions such that the four halides of Fig. 4.1 (p. 82) all react *via* the S_N1 pathway.

Thus, as the S_N2 rate is expected to *decrease*, and the S_N1 rate to *increase*, across the series in Fig. 4.1, the reason for the observed pattern of reaction rates, and changeover in reaction pathway, becomes apparent.

A similar mechanistic changeover is observed, though considerably sooner, in traversing the series:



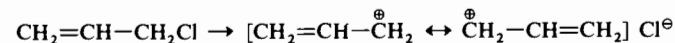
Thus for hydrolysis in 50% aqueous acetone, a mixed second and first order rate equation is observed for phenylchloromethane (benzyl chloride, 10)—moving over almost completely to the S_N1 mode in water alone. Diphenylchloromethane (11) is found to follow a first order rate equation, with a very large increase in total rate, while with triphenylchloromethane (trityl chloride, 12) the ionisation is so pronounced that the compound exhibits electrical conductivity when dissolved in liquid SO_2 . The main reason for the greater promotion of ionisation—with consequent earlier changeover to the S_N1 pathway in this series—is the considerable stabilisation of the carbocation, by delocalisation of its positive charge, that is now possible:



This is a classical example of an ion stabilised by charge delocalisation *via* the agency of the delocalised π orbitals of the benzene nucleus (*cf.* the negatively charged phenoxide ion, p. 23). The effect will become progressively more pronounced, and S_N1 attack further facilitated, with $(C_6H_5)_2CHCl$ (11) and $(C_6H_5)_3CCl$ (12), as the possibilities for delocalising the positive charge are increased in the carbocations to which these latter halides give rise.

S_N2 attack on the CH_2 in (10) is found to proceed at very much the same rate as on that in $MeCH_2Cl$, suggesting that any adverse steric crowding in the T.S. by the bulky C_6H_5 group is compensated by a small electronic (inductive?) effect promoting reaction.

Similar carbocation stabilisation can also occur in the hydrolysis of allyl halides, e.g. 3-chloropropene:

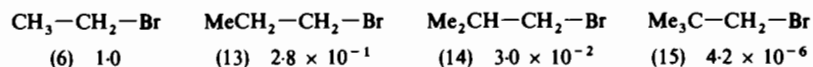


S_N1 attack is thus promoted and allyl, like benzyl, halides are normally more reactive than species, e.g. $CH_3CH_2CH_2Cl$ and $C_6H_5CH_2CH_2CH_2Cl$, in which such carbocation stabilisation cannot take place. S_N2 attack is also speeded up, compared with $CH_3CH_2CH_2Cl$, presumably because any electronic effect of the double bond—promoting reaction—is not here nullified by an adverse steric effect, as with the bulky C_6H_5 group in $C_6H_5CH_2Cl$ (*cf.* above). The proportion of the total reaction proceeding by each of the two pathways is found to depend on the conditions: more powerful nucleophiles promoting the S_N2 mode (*cf.* p. 96).

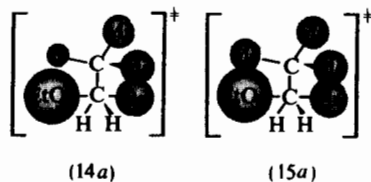
By contrast, vinyl halides such as chloroethene, $CH_2=CHCl$, and halogenobenzenes are very unreactive towards nucleophiles. This stems from the fact that the halogen atom is now bonded to an sp^2 hybridised carbon, with the result that the electron pair of the $C-Cl$ bond is drawn closer to carbon than in the bond to an sp^3 hybridised carbon. The $C-Cl$ is found to be stronger, and thus less easily broken, than in, for example, CH_3CH_2Cl , and the $C-Cl$ dipole is smaller; there is thus less tendency to ionisation (S_N1) and a less positive carbon for OH^- to attack (S_N2); the π electrons of the double bond also inhibit the close approach of an attacking nucleophile. The double bond would not help to stabilise either the S_N2 transition state or the carbocation involved in the S_N1 pathway. Very much the same considerations apply to halogenobenzenes, with their sp^2 hybridised carbons and the π orbital system of the benzene nucleus; their reactions, which though often bimolecular are not in fact simply S_N2 in nature, are discussed further below (p. 170).

The influence of steric factors on the reaction pathway is particularly observed when substitution takes place at the β -position. Thus for the

series,

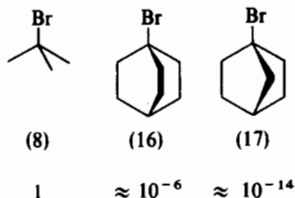


the figures quoted are relative rates of reaction ($\text{S}_{\text{N}}2$ throughout) with $\text{EtO}^\ominus$ in EtOH at 55° . Any differences in electronic effect of the Me groups through two saturated carbon atoms would be very small, and the reason for the rate differences is steric: increased difficulty of approach of $\text{EtO}^\ominus$ 'from the back' of the carbon atom carrying Br, and increased crowding in the resultant T.S. The reason for the particularly large drop in rate between 1-bromo-2-methylpropane (14) and 1-bromo-2,2-dimethylpropane (neopentyl bromide, 15) is that the T.S. for the former, though somewhat crowded, can, by rotation about the $\text{C}_\alpha-\text{C}_\beta$ bond, adopt one conformation (14a) in which the attacking $\text{EtO}^\ominus$ is interfered with only by H, while no such relief of crowding is open in the T.S. (15a) for the latter (but see also, p. 110):



The T.S. (15a) will thus be at a much higher energy level, $\Delta G^\ddagger$ (p. 38) will be larger and the reaction rate correspondingly lower.

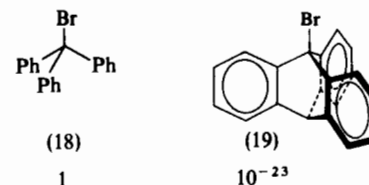
The effect of structure on relative reactivity may be seen particularly clearly when a halogen atom is located at the bridgehead of a bicyclic system. Thus the following rates were observed for solvolysis in 80% aqueous ethanol at 25° :



All are tertiary halides so that attack by the $\text{S}_{\text{N}}2$ mode would not be expected to occur on (16) or (17) any more than it did on (8) (cf. p. 82). $\text{S}_{\text{N}}2$ attack 'from the back' on the carbon atom carrying Br would in any case be prevented in (16) and (17) both sterically by their cage-like structure, and also by the impossibility of forcing their fairly rigid framework through transition states with the required planar distribution of bonds to the bridgehead carbon atom (cf. p. 84). Solvolysis *via* rate-limiting formation of the ion pair ($\text{S}_{\text{N}}1$), as happens with (8) is

also inhibited because the resultant carbocations from (16) and (17) would be unable, because of their rigid frameworks, to stabilise themselves by collapsing to the stable planar state. These carbocation intermediates are thus of very much higher energy level than usual, and therefore are formed only slowly and with reluctance. The very greatly reduced solvolysis rate of (17) compared with (16) reflects the greater rigidity about the bridgehead (cationic) carbon with a one-carbon (17), than with a two-carbon (16), bridge.

This rigidity is carried even further in 1-bromotriptycene (19),

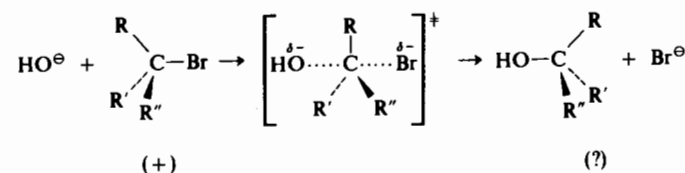


in which the bromine atom is found to be virtually inert to nucleophiles. Despite the formal resemblance in the environment of the bromine atom in (19) to that in (18), they are found to differ in their rate of reaction under parallel conditions by a factor of $\approx 10^{-23}$:!! This is because stabilisation of the carbocation from (18) can occur by delocalisation of its charge through the π orbital systems of the three benzene rings; whereas the extremely rigid structure of (19) will hold the cation's empty orbital (from loss of $\text{Br}^\ominus$) all but at right angles to these π orbital systems, thus preventing such delocalisation.

4.4 STEREOCHEMICAL IMPLICATIONS OF MECHANISM

Hydrolysis of an optically active form of a *chiral** halide presents some interesting stereochemical features. Thus considering each pathway in turn:

4.4.1 $\text{S}_{\text{N}}2$ mechanism: inversion of configuration



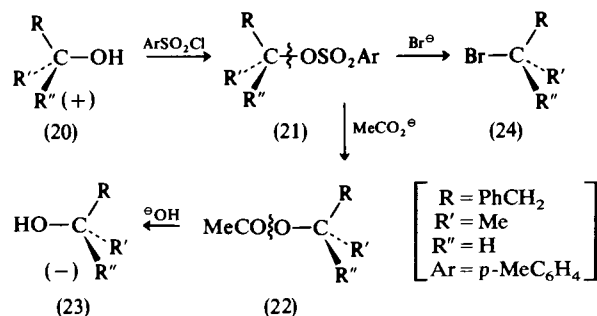
* A *chiral* compound is one that is not superimposable on its mirror image.

It will be seen that the spatial arrangement of the three residual groups attached to the carbon atom attacked has been effectively turned inside out. The carbon atom is said to have undergone *inversion* of its *configuration* (the arrangement in space of the groups attached to it). Indeed, if the product could be the bromide, instead of, as here, the corresponding alcohol, it would be found to rotate the plane of polarisation of plane polarised light in the opposite direction, i.e. (–), to the starting material, (+), for it would, of course, be its mirror image (*cf.* p. 89). The actual product is the alcohol, however, and we are unfortunately not able to tell, merely by observing its direction of optical rotation, whether it has the same or the opposite configuration to the bromide from which it was derived: compounds, other than mirror images, that have opposite configurations do not necessarily exhibit opposite directions of optical rotation, while compounds that have the same configuration do not necessarily exhibit the same direction of optical rotation. Thus in order to confirm that the above S_N2 reaction is, in practice, attended by an inversion of configuration, as theory requires, it is necessary to have an independent method for relating the configuration of starting material and product, e.g. the bromide and corresponding alcohol above.

4.4.2 Determination of relative configuration

This turns essentially on the fact that if a chiral compound undergoes a reaction in which a bond joining one of the groups to the chiral centre is broken, then the centre may—though it need not of necessity—undergo inversion of configuration; while if the compound undergoes reaction in which no such bond is broken then the chiral centre will preserve its configuration intact.

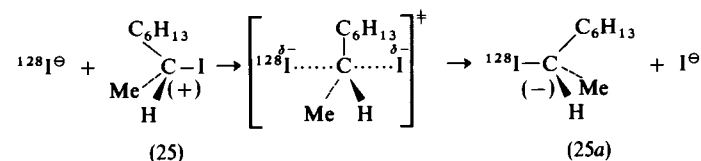
Thus in the series of reactions on the optically active(+) alcohol (20),



formation of an ester with 4-methylbenzenesulphonyl(tosyl) chloride

is known not to break the C—O bond of the alcohol,* hence the tosylate (21) must have the same configuration as the original alcohol. Reaction of this ester (21) with $\text{MeCO}_2^\ominus$ is known to be a displacement in which $\text{ArSO}_3^\ominus$ ($\text{Ar} = p\text{-MeC}_6\text{H}_4$) is expelled and $\text{MeCO}_2^\ominus$ introduced,* hence the C—O bond is broken in this reaction, and inversion of configuration can thus take place in forming the acetate (22). Alkaline hydrolysis of the acetate ($22 \rightarrow 23$) can be shown not to involve fission of the alkyl-oxygen C—O linkage,† so the alcohol (23) must have the same configuration as the acetate (22). As (23) is found to be the mirror image of the starting material (20)—opposite direction of optical rotation—an inversion of configuration must have taken place during the series of reactions, and can have occurred only during reaction of $\text{MeCO}_2^\ominus$ with the tosylate (21). Reaction of this tosylate (21) with a number of nucleophiles showed that inversion of configuration occurred in each case: it may thus be concluded with some confidence that it occurs on reaction with $\text{Br}^\ominus$ to yield the bromide (24), i.e. that the bromide (24), like the acetate (22), has the opposite configuration to the original alcohol (20).

The general principle—that bimolecular (S_N2) displacement reactions are attended by inversion of configuration—was established in an elegant and highly ingenious experiment, in which an optically active alkyl halide undergoes displacement by the same—though isotopically labelled—halide ion as nucleophile, e.g. radioactive $^{128}\text{I}^\ominus$ on (+)-2-iodooctane (25):



The displacement was monitored by observing the changing distribution of radioactive ^{128}I between the inorganic (sodium) iodide and 2-iodooctane, and it was found, under these conditions, to be second order overall (first order with respect to $^{128}\text{I}^\ominus$ and to 2-iodooctane) with $k_2 = 3.00 \pm 0.25 \times 10^{-5}$ (at 30°).

If *inversion* takes place, as S_N2 requires, the optical activity of the solution will decline to zero, i.e. *racemisation* will occur. This will happen because inversion of the configuration of a molecule of (+) (25) results in formation of a molecule of its mirror image (–)

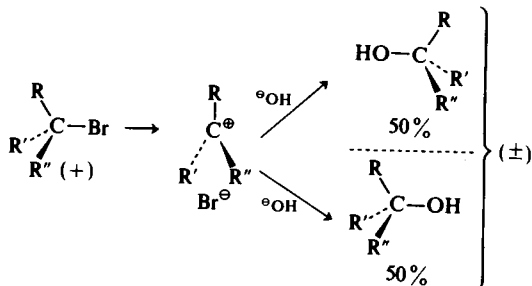
* That such is the case may be shown by using an alcohol labelled with ^{18}O in its OH group, and demonstrating that this atom is not eliminated on forming the tosylate; it is, however, eliminated when the tosylate is reacted with $\text{MeCO}_2^\ominus$.

† Hydrolysis of an acetate in which the alcohol-oxygen atom is ^{18}O labelled fails to result in the latter's replacement, thus showing that the alkyl-oxygen bond of the acetate is not broken during its hydrolysis (*cf.* p. 47).

(25a), which 'pairs off' with a second molecule of (+) (25) to form a ($\pm$) racemate: the observed rate of *racemisation* will thus be twice the rate of inversion. The reaction was monitored polarimetrically, the rate of racemisation measured thereby, and the rate of inversion calculated from it: it was found to have $k = 2.88 \pm 0.03 \times 10^{-5}$ (at 30°).

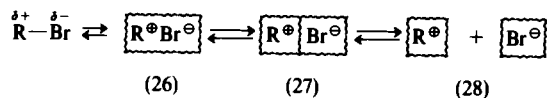
The rate of displacement and of inversion are thus identical within the limits of experimental error, and it thus follows that each act of bimolecular displacement must thus proceed with inversion of configuration. Having shown that S_N2 reactions are attended by inversion of configuration, independent demonstration that a particular reaction occurs *via* the S_N2 mode is often used to correlate the configuration of product and starting material in the reaction.

4.4.3 S_N1 mechanism: racemisation?



As the carbocation formed in the slow, rate-limiting step of the reaction is planar, it might be expected that subsequent attack by a nucleophile such as $^{\ominus}\text{OH}$, or the solvent (H_2O), would take place with equal readiness from either side of this planar carbocation; leading to a 50/50 mixture of species having the same, and the opposite, configuration as the starting material, i.e. that *racemisation* would take place yielding an optically inactive ($\pm$) product.

In practice, however, the expected racemisation—and nothing but racemisation—is rarely observed, it is almost always accompanied by some degree of inversion. The relative proportions of the two are found to depend on: (a) the structure of the halide, in particular the relative stability of the carbocation to which it can give rise; and (b) the solvent, in particular on its ability as a nucleophile. The more stable the carbocation, the greater is the proportion of racemisation; the more nucleophilic the solvent, the greater is the proportion of inversion. These observations become understandable if the rate-limiting S_N1 ionisation follows the sequence:



Here (26) is an *intimate* ion pair in which the jointly solvated gegenions are in very close association with no solvent molecules between them, (27) is a *solvent-separated* ion pair, and (28) represents the now dissociated, and separately solvated, pair of ions.

In a solvolysis reaction, attack on $\text{R}^{\oplus}$ by a solvent molecule, e.g. H_2O , in (26) is likely to lead to inversion, as attack can take place (by the solvent envelope) on the 'back' side of $\text{R}^{\oplus}$, but not on the 'front' side where there are no solvent molecules, and which is shielded by the $\text{Br}^{\ominus}$ gegen ion. Attack in (27) is more likely to lead to attack from either side, leading to racemisation, while attack on (28) can clearly happen with equal facility from either side. Thus the longer the life of $\text{R}^{\oplus}$, i.e. the longer it escapes nucleophilic attack, the greater the proportion of racemisation that we should expect to occur. The life of $\text{R}^{\oplus}$ is likely to be longer the more stable it is—(a) above—but the shorter the more powerfully nucleophilic the solvent—(b) above.

Thus solvolysis of $(+)\text{C}_6\text{H}_5\text{CHMeCl}$, which can form a stabilised benzyl type carbocation (cf. p. 84), leads to 98% racemisation while $(+)\text{C}_6\text{H}_{13}\text{CHMeCl}$, where no comparable stabilisation can occur, leads to only 34% racemisation. Solvolysis of $(+)\text{C}_6\text{H}_5\text{CHMeCl}$ in 80% acetone/20% water leads to 98% racemisation (above), but in the more nucleophilic water alone to only 80% racemisation. The same general considerations apply to nucleophilic displacement reactions by Nu^- as to solvolysis, except that $\text{R}^{\oplus}$ may persist a little further along the sequence because part at least of the solvent envelope has to be stripped away before Nu^- can get at $\text{R}^{\oplus}$. It is important to notice that racemisation is clearly very much less of a stereochemical requirement for S_N1 reactions than inversion was for S_N2 .

4.4.4 The mechanistic borderline

Reference has already been made (p. 82) to the fact that the reactions of some substrates, e.g. secondary halides, may follow a mixed first/second order rate equation. The question then arises whether such a reaction is proceeding *via* both S_N2 and S_N1 pathways simultaneously (their relative proportions depending on the solvent, etc.) or whether it is proceeding *via* some specific, 'in between' mechanistic pathway.

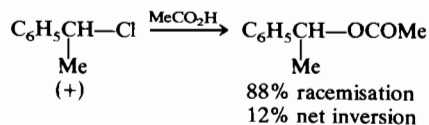
In solvolytic reactions like those we have just been considering, where the solvent itself is the nucleophile, such mixed kinetics may not be detectable, irrespective of what is actually happening, as both S_N1 and S_N2 pathways are likely to follow a rate equation of the form:

$$\text{Rate} = k[\text{R}-\text{X}]$$

This is so because in the S_N2 pathway the concentration of nucleophile will remain essentially constant throughout the reaction as—being also the solvent—it is present in very large, unchanging

excess. This raises the question whether the mixture of racemisation/inversion observed in such cases stems from the simultaneous operation of S_N1 and S_N2 pathways for solvolysis, rather than *via* the relatively elaborate, variable ion pair hypothesis advanced above.

In some cases at least it is possible to demonstrate that a 'mixed' $S_N1 + S_N2$ pathway is *not* operating. Thus solvolysis of the halide, $(+)\text{C}_6\text{H}_5\text{CHMeCl}$, mentioned above, but this time in MeCO_2H ,

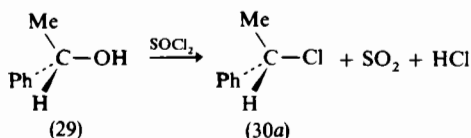


was found to lead to 88% racemisation, and 12% net inversion. Adding the much more powerfully nucleophilic MeCO_2^- (as $\text{MeCO}_2^-\text{Na}^+$) to the reaction mixture was found to result in: (a) no increase in the overall reaction rate, and (b) no increase in the proportion of net inversion. This strongly suggests that the inversion that is observed does *not* stem from part of the overall reaction proceeding *via* an S_N2 pathway simultaneously with the (major) S_N1 mode. If it did, we would expect the change to a much more powerful nucleophile ($\text{MeCO}_2\text{H} \rightarrow \text{MeCO}_2^-$) to lead to marked increases in both (a) and (b) above.

A good deal of interest, and controversy, has centred on whether in the last analysis there is perhaps a continuous spectrum of mechanistic pathways intermediate between S_N2 and S_N1 : these imperceptibly shading into each other *via* gradually varying transition states from the pure S_N2 side, and *via* gradually varying ion pair/solvent combinations from the pure S_N1 side. It is an area in which theory has shaded over into semantics if, indeed, not even into theology!

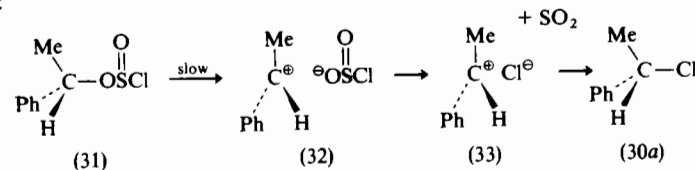
4.4.5 S_Ni mechanism: retention of configuration

Despite what has been said above about displacement reactions leading to inversion of configuration, to racemisation, or to a mixture of both, a number of cases are known of reactions that proceed with actual retention of configuration, i.e. in which the starting material and product have the same configuration. One reaction in which this has been shown to occur is in the replacement of OH by Cl through the use of thionyl chloride, SOCl_2 :



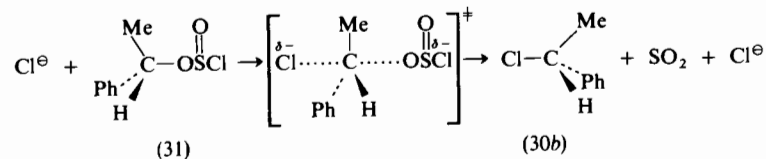
The reaction has been shown to follow a second order rate equation, $\text{rate} = k_2[\text{ROH}][\text{SOCl}_2]$, but clearly cannot proceed by the simple S_N2 mode for this would lead to inversion of configuration (p. 87) in the product, which is not observed.

Carrying out the reaction under milder conditions allows of the isolation of an alkyl chlorosulphite, ROSOCl (31), and this can be shown to be a true intermediate. The chlorosulphite is formed with retention of configuration, the $\text{R}-\text{O}$ bond not being broken during the reaction. The rate at which the alkyl chlorosulphite intermediate (31) breaks down to the product, RCl (30a), is found to increase with increasing polarity of the solvent, and also with increasing stability of the carbocation R^+ : an ion pair, R^+OSOCl^- (32), is almost certainly involved. Provided collapse of the ion pair to products then occurs rapidly, i.e. in the intimate ion pair (33) within a solvent cage (*cf.* p. 90), then attack by Cl^- is likely to occur on the same side of R^+ from which $^-\text{OSOCl}$ departed, i.e. with retention of configuration:



Whether the breaking of the $\text{C}-\text{O}$ and the $\text{S}-\text{Cl}$ bonds occurs simultaneously, or whether the former occurs first, is still a matter of debate.

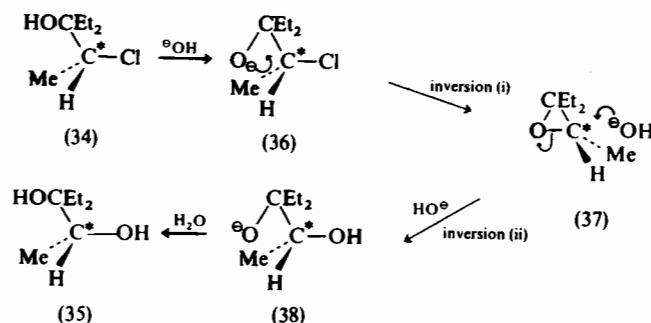
It is interesting that if the SOCl_2 reaction on ROH (29) is carried out in the presence of pyridine, the product RCl is found now to have undergone inversion of configuration (30b). This occurs because the HCl produced during the formation of (31) from ROH and SOCl_2 is converted by pyridine into $\text{C}_5\text{H}_5\text{NH}^+\text{Cl}^-$ and Cl^- , being an effective nucleophile, attacks (31) 'from the back' in a normal S_N2 reaction with inversion of configuration:



4.4.6 Neighbouring group participation: 'retention'

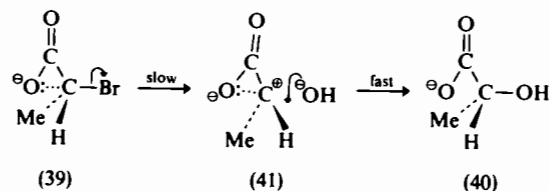
There are also some examples of retention of configuration in nucleophilic displacement reactions where the common feature is an atom or group—close to the carbon undergoing attack—which has an electron

pair available. This *neighbouring group* can use its electron pair to interact with the 'backside' of the carbon atom undergoing substitution, thus preventing attack by the nucleophilic reagent; attack can thus take place only 'from the frontside', leading to retention of configuration. Thus base hydrolysis of the 1,2-chlorohydrin (34) is found to yield the 1,2-diol (35) with the same configuration (retention):



Initial attack by base on (34) yields the alkoxide anion (36), *internal* attack by this $\text{RO}^{\ominus}$ then yields the epoxide (37) with inversion of configuration at C^* (these cyclic intermediates can actually be isolated in many cases); this carbon atom[†], in turn, undergoes ordinary $\text{S}_{\text{N}}2$ attack by $^{\ominus}\text{OH}$, with a second inversion of configuration at C^* . Finally, this second alkoxide anion (38) abstracts a proton from the solvent to yield the product 1,2-diol (35) with the same configuration as the starting material (34). This *apparent* retention of configuration has, however, been brought about by two successive inversions.

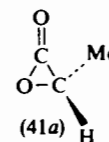
Another example of oxygen as a neighbouring group occurs in the hydrolysis of the 2-bromopropanoate anion (39) at low $[^{\ominus}\text{OH}]$, which is also found to proceed with retention of configuration (40). The rate is found to be independent of $[^{\ominus}\text{OH}]$, and the reaction is believed to proceed:



Whether the intermediate (41) is a zwitterion as shown or a highly

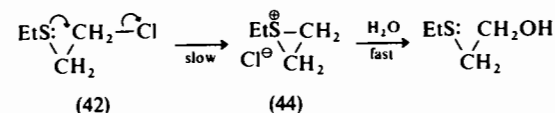
[†] Preferential attack takes place on this, rather than the other, carbon of the three-membered ring as it will be the more positive of the two, carrying as it does only one electron-donating alkyl group.

labile α -lactone (41a)

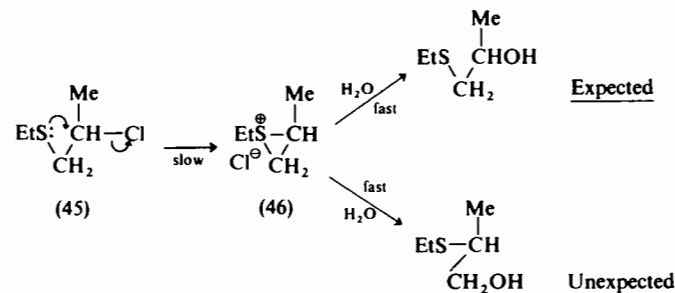


has not been clearly established. As the concentration of nucleophile, $[^{\ominus}\text{OH}]$, is increased an increasing proportion of normal $\text{S}_{\text{N}}2$ 'attack from the back', with inversion of configuration, is observed.

Neighbouring group effects have also been observed with atoms other than oxygen, e.g. sulphur and nitrogen, and in situations where, though no stereochemical point is at issue, unexpectedly rapid rates suggest a change in reaction pathway. Thus $\text{EtSCH}_2\text{CH}_2\text{Cl}$ (42) is found to undergo hydrolysis 10^4 times faster than $\text{EtOCH}_2\text{CH}_2\text{Cl}$ (43) under comparable conditions, and this has been interpreted as involving S: acting as a neighbouring group:



By contrast, O: in (43) is sufficiently electronegative not to donate an electron pair (unlike $\text{O}^{\ominus}$ in $\text{RO}^{\ominus}$ and $\text{RCO}_2^{\ominus}$ above), and hydrolysis of $\text{EtOCH}_2\text{CH}_2\text{Cl}$ thus proceeds *via* ordinary $\text{S}_{\text{N}}2$ attack by an external nucleophile—which is likely to be very much slower than the *internal* nucleophilic attack in (42) $\rightarrow$ (44). That a cyclic sulphonium salt such as (44) is involved is demonstrated by the hydrolysis of the analogue (45), which yields two alcohols (the unexpected one in greater yield) indicating the participation of the unsymmetrical intermediate (46):



N: can act as a neighbouring group in similar circumstances, e.g. the hydrolysis of $\text{Me}_2\text{NCH}_2\text{CH}_2\text{Cl}$, but the rate is markedly slower,